Improvement in Food Resources

The big idea

India's growing population needs more food without destroying the environment. Scientists improve **crops** and **animals** so that we produce more, better-quality food from the same land: this is the idea behind the Green Revolution (crops) and White Revolution (milk).

Crop variety improvement

Crops are bred by **hybridisation** (crossing varieties) or by introducing a gene (**genetically modified** crops) to get:

- higher yield,
- better quality (protein in pulses, oil in oilseeds),
- resistance to diseases and pests,
- tolerance to heat, drought, salinity,
- wider adaptability and suitable agronomic traits (shorter, dwarf plants for less lodging).

Seasons: **kharif** crops grow in the rainy season, June–October (paddy, soyabean, maize, cotton). **Rabi** crops grow in winter, November–April (wheat, gram, peas, mustard).

Crop production management

- **Nutrients**: plants need 16 nutrients. Macronutrients (N, P, K, Ca, Mg, S) are needed in large amounts; micronutrients (Fe, Mn, Zn, Cu, B, Mo, Cl) in small amounts.

- **Manure** (organic, improves soil structure) vs **fertilisers** (chemical, quick but can harm soil and water if overused). **Organic farming** uses minimal chemicals.
- **Irrigation**: wells, canals, river-lift systems, tanks, rainwater harvesting.
- **Cropping patterns**: **mixed cropping** (two crops together in the same field), **intercropping** (in a set row pattern), **crop rotation** (different crops in sequence, e.g. a legume to restore nitrogen).

Crop protection

Weeds, insect pests and diseases reduce yield. Control them with pesticides used carefully, resistant varieties, timely sowing, summer ploughing and proper storage. Grain is protected from moisture, insects, rodents and fungi by cleaning, drying and fumigation.

Animal husbandry

- **Cattle farming**: milk-producing animals (milch) and draught animals. Exotic breeds (Jersey, Brown Swiss) give long lactation; local breeds (Red Sindhi, Sahiwal) resist disease; crossbreeding combines both. Proper shelter, cleanliness, balanced feed (roughage + concentrates) and vaccination keep them healthy.
- **Poultry**: layers (eggs) and broilers (meat); crossbreeding of Indian (Aseel) and foreign (Leghorn) breeds.
- **Fish production**: capture fishing (marine, inland) and **culture** fishery (aquaculture). **Composite fish culture** puts 5–6 species that feed at different levels in one pond (catla at the surface, rohu in the middle, mrigal at the bottom).
- **Bee-keeping (apiculture)**: honey and wax; Italian bee *Apis mellifera* gives high yield. Pasturage (flowers available) decides honey quality.

Quick recap

- Improve crops by hybridisation and GM for yield, quality and resistance.
- Kharif (rainy season) and rabi (winter) crops.
- Manage nutrients, irrigation and cropping patterns; protect crops and stored grain.
- Animal husbandry: cattle, poultry, fish and bee-keeping.

Practice questions

1. Which is a rabi crop?
   (a) Paddy (b) Maize (c) Wheat (d) Cotton
2. Growing two or more crops in a definite row pattern is
   (a) mixed cropping (b) intercropping (c) crop rotation (d) monoculture
3. Which is a macronutrient for plants?
   (a) Iron (b) Zinc (c) Nitrogen (d) Boron
4. In composite fish culture, the fish that feeds at the bottom of the pond is
   (a) catla (b) rohu (c) mrigal (d) grass carp
5. Which bee variety is commonly used for high honey yield?
   (a) Apis mellifera (Italian bee) (b) Apis dorsata (c) Apis florae (d) Apis cerana only

Answers: 1. c 2. b 3. c 4. c 5. a